Problem 16.3.15

Evaluate counterclockwise.

$$\oint_C \tan 2\pi z \; dz, \quad C : |z - 0.2| = 0.2$$

Solution

The function $f(z) = \tan 2\pi z = \frac{\sin 2\pi z}{\cos 2\pi z}$ has simple poles at

$$\cos 2\pi z = 0 \implies 2\pi z = \frac{\pi}{2} + n\pi \implies z = \frac{1}{4} + \frac{n}{2}, \; n \in \mathbb{Z}$$

Only $z_0 = \frac{1}{4}$ lies in C,

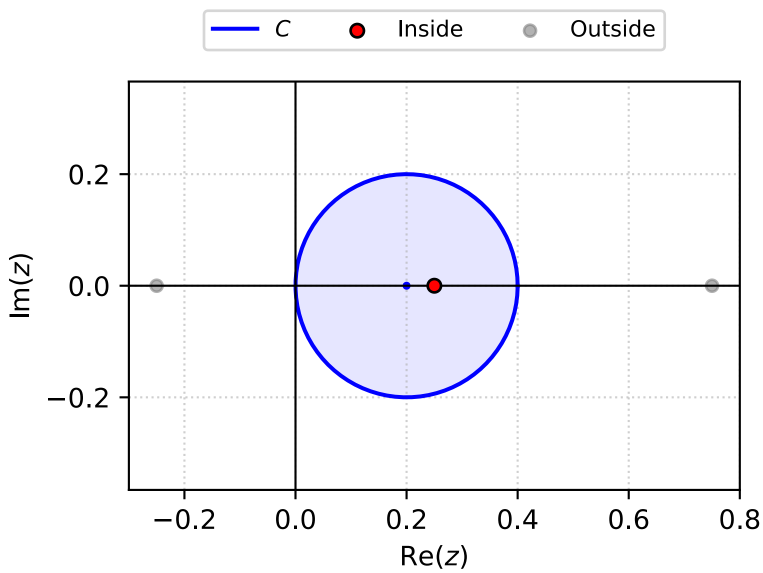

Figure 1: Poles

$$\operatorname*{Res}_{z=z_0} f(z) = \frac{p(z_0)}{q'(z_0)} = \frac{\sin 2\pi z}{-2\pi \sin 2\pi z} = -\frac{1}{2\pi}$$

The integral is then in the form,

$$\oint_C f(z) \; dz = 2\pi i \operatorname*{Res}_{z=z_0} f(z)$$

$$\oint_C \tan 2\pi z \; dz = 2\pi i \operatorname*{Res}_{z=\frac{1}{4}} f(z) = \boxed{-i}$$

Problem 16.3.17

Evaluate counterclockwise.

$$\oint_C \frac{e^z}{\cos z}\, dz, \quad C : \left| z - \frac{\pi i}{2} \right| = 4.5$$

Solution

The function $f(z) = \frac{e^z}{\cos z}$ has simple poles at

$$\cos z = 0 \implies z = \frac{\pi}{2} + n\pi,\ n \in \mathbb{Z}$$

Poles at $z_{1,2} = \pm\frac{\pi}{2}$ lie within the region,

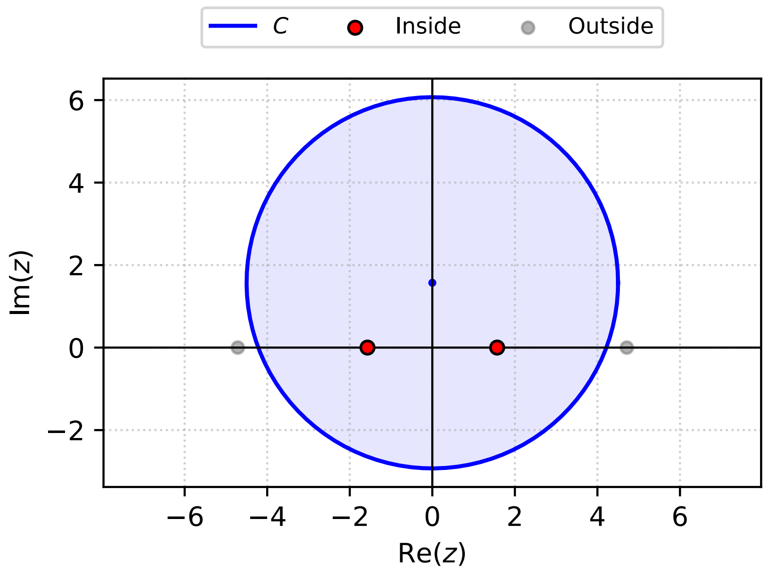

Figure 2: Poles

$$\operatorname*{Res}_{z=z_0} f(z) = \frac{p(z_0)}{q'(z_0)} = -\frac{e^{z_0}}{\sin z_0}$$

$$\operatorname*{Res}_{z=\frac{\pi}{2}} f(z) = -\frac{e^{\frac{\pi}{2}}}{\sin \frac{\pi}{2}} = -e^{\frac{\pi}{2}}$$

$$\operatorname*{Res}_{z=-\frac{\pi}{2}} f(z) = -\frac{e^{-\frac{\pi}{2}}}{\sin -\frac{\pi}{2}} = e^{-\frac{\pi}{2}}$$

The integral is then in the form,

$$\oint_C f(z)\, dz = 2\pi i \sum_{j=1}^{k} \operatorname*{Res}_{z=z_j} f(z)$$

$$\oint_C \frac{e^z}{\cos z}\, dz = 2\pi i \left(-e^{\frac{\pi}{2}} + e^{-\frac{\pi}{2}}\right) = \boxed{-4\pi i \sinh\left(\frac{\pi}{2}\right)}$$

Problem 16.3.19

Evaluate counterclockwise.

$$\oint_C \frac{\sinh z}{2z - i}\, dz, \quad C : |z - 2i| = 2$$

Solution

The function $f(z) = \frac{\sinh z}{2z-i}$ has simple poles at

$$2z - i = 0 \implies z = \frac{i}{2}$$

The pole at $z_0 = \frac{i}{2}$ lies within the region,

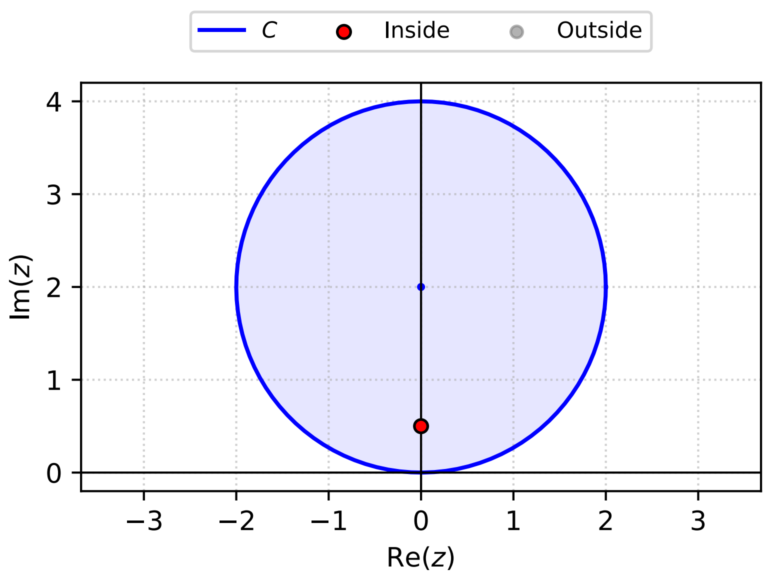

Figure 3: Poles

$$\operatorname*{Res}_{z=z_0} f(z) = \frac{p(z_0)}{q'(z_0)} = \frac{1}{2}\sinh z_0$$

$$\operatorname*{Res}_{z=\frac{i}{2}} f(z) = \frac{1}{2}\sinh\frac{i}{2} = \frac{i}{2}\sin\frac{1}{2}$$

The integral is then in the form,

$$\oint_C f(z)\, dz = 2\pi i \operatorname*{Res}_{z=z_0} f(z)$$

$$\oint_C \frac{\sinh z}{2z - i}\, dz = 2\pi i\left(\frac{i}{2}\sin\frac{1}{2}\right) = \boxed{-\pi\sin\frac{i}{2}}$$

Problem 16.3.21

Evaluate counterclockwise.

$$\oint_C \frac{\cos \pi z}{z^5}\, dz, \quad C : |z| = \frac{1}{2}$$

Solution

The function $f(z) = \frac{\cos \pi z}{z^5}$ has a 5th-order pole at

$$z^5 = 0 \implies z = 0$$

The pole at $z_0 = 0$ lies within the region,

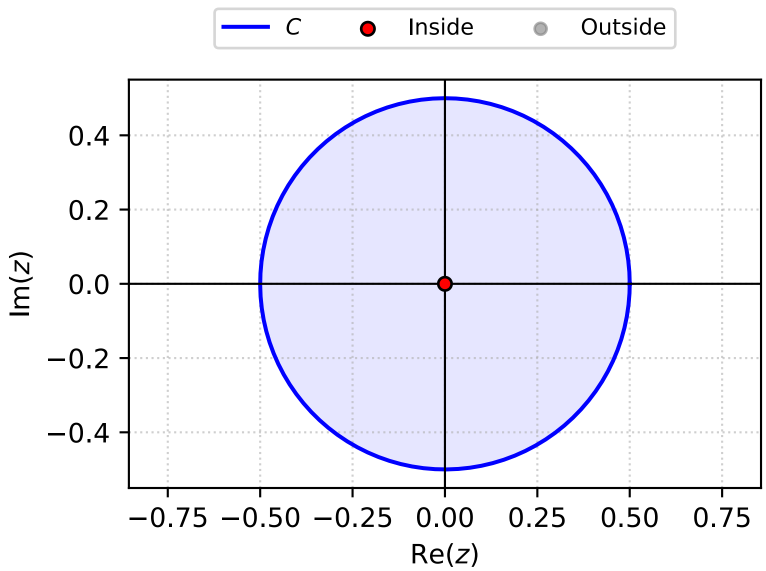

Figure 4: Poles

$$\operatorname*{Res}_{z=0} f(z) = \frac{1}{4!} \lim_{z \to 0} \left\{ \frac{d^4}{dz^4} \left[z^4 f(z) \right] \right\} = \frac{1}{24} \frac{d^4}{dz^4} \left[\cos \pi z \right] \Big|_{z=0} = \frac{\pi^4}{24} \cos \pi z \Big|_{z=0} = \frac{\pi^4}{24}$$

The integral is then in the form,

$$\oint_C f(z)\, dz = 2\pi i \operatorname*{Res}_{z=z_0} f(z)$$

$$\oint_C \frac{\cos \pi z}{z^5}\, dz = 2\pi i \left(\frac{\pi^4}{24} \right) = \boxed{\frac{i\pi^5}{12}}$$

Problem 16.4.1

Evaluate.

$$\int_0^{\pi} \frac{2}{k - \cos\theta}\, d\theta$$

Solution

The integral can be converted to a complex integral by substituting $z = e^{i\theta}$

$$\int_0^{\pi} \frac{2}{k - \cos\theta}\, d\theta = \int_0^{2\pi} \frac{1}{k - \cos\theta}\, d\theta = \oint_C \frac{1}{k - \frac{1}{2}\left(z + z^{-1}\right)} \frac{dz}{iz} = 2i \oint_C \frac{1}{z^2 - 2kz + 1}\, dz$$

Where C is the the unit circle in the complex plane.

$f(z) = \frac{1}{z^2 - 2kz + 1}$ has simple poles where

$$z^2 - 2kz + 1 = 0 \implies z = \frac{2k \pm \sqrt{4k^2 - 4}}{2} = k \pm \sqrt{k^2 - 1}$$

$$z_1 = k - \sqrt{k^2 - 1} \quad z_2 = k + \sqrt{k^2 - 1}$$

$$\operatorname*{Res}_{z=z_0} f(z) = \frac{1}{2z_0 - 2k} = \frac{1}{2(z_0 - k)}$$

$$\operatorname*{Res}_{z=z_{1,2}} f(z) = \frac{1}{2(k \pm \sqrt{k^2 - 1} - k)} = \pm\frac{1}{2\sqrt{k^2 - 1}}$$

Assuming $k > 1$, only z_1 lies within C,

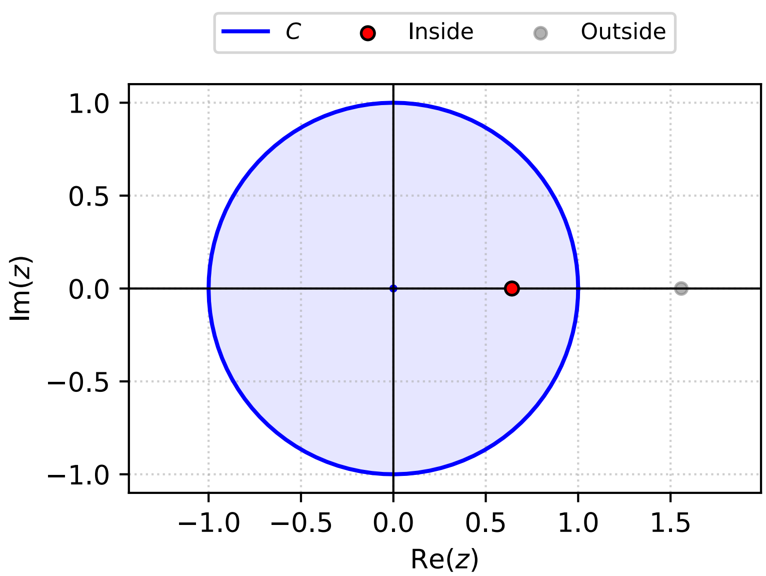

Figure 5: Poles (e.g. $k = 1.1$)

$$
\begin{aligned}
\oint_C f(z)\ dz &= 2\pi i \operatorname*{Res}_{z=z_0} f(z) \\
\oint_C \frac{1}{z^2 - 2kz + 1} &= 2\pi i \left(-\frac{1}{2\sqrt{k^2-1}} \right) \quad = -\frac{\pi i}{\sqrt{k^2-1}} \\
\int_0^\pi \frac{2}{k - \cos\theta}\ d\theta &= 2i \left(-\frac{\pi i}{\sqrt{k^2-1}} \right) \\
&= \boxed{\frac{2\pi}{\sqrt{k^2-1}}}
\end{aligned}
$$

Problem 16.4.7

Evaluate.

$$\int_0^{2\pi} \frac{a}{a - \sin\theta}\, d\theta$$

Solution

The integral can be converted to a complex integral by substituting $z = e^{i\theta}$

$$\int_0^{2\pi} \frac{a}{a - \sin\theta}\, d\theta = \oint_C \frac{a}{a - \frac{1}{2i}\left(z - z^{-1}\right)} \frac{dz}{iz} = -2a \oint_C \frac{1}{z^2 - 2aiz - 1}\, dz$$

Where C is the the unit circle in the complex plane.
$f(z) = \frac{1}{z^2 - 2aiz - 1}$ has simple poles where

$$z^2 - 2aiz - 1 = 0 \implies z = \frac{2ai \pm \sqrt{-4a^2 + 4}}{2} = ai \pm \sqrt{1 - a^2}$$

Assuming $a > 1$

$$z_1 = \left(a - \sqrt{a^2 - 1}\right) i \quad z_2 = \left(a + \sqrt{a^2 - 1}\right) i$$

$$\operatorname*{Res}_{z=z_0} f(z) = \frac{1}{2z_0 - 2ai} = \frac{1}{2(z_0 - ai)}$$

$$\operatorname*{Res}_{z=z_{1,2}} f(z) = \frac{1}{2(\left(a \pm \sqrt{a^2 - 1}\right) i - ai)} = \pm \frac{1}{2i\sqrt{a^2 - 1}}$$

Only z_1 lies within C,

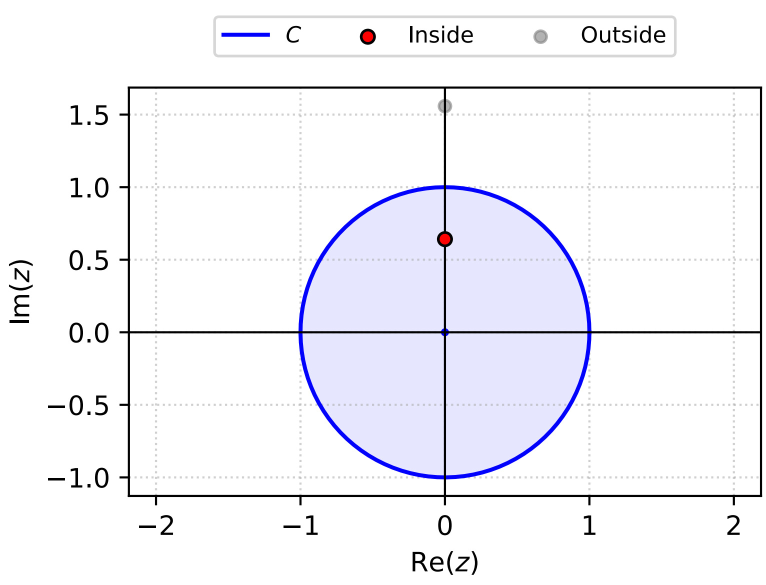

Figure 6: Poles (e.g. $a = 1.1$)

$$\begin{aligned}
\oint_C f(z)\ dz &= 2\pi i \operatorname*{Res}_{z=z_0} f(z) \\
\oint_C \frac{1}{z^2 - 2kz + 1} &= 2\pi i \left(-\frac{1}{2i\sqrt{a^2-1}} \right) \quad = -\frac{\pi}{\sqrt{a^2-1}} \\
\int_0^{2\pi} \frac{a}{a - \sin\theta}\ d\theta &= -2a \left(-\frac{\pi}{\sqrt{a^2-1}} \right) \\
&= \boxed{\frac{2a\pi}{\sqrt{a^2-1}}}
\end{aligned}$$

Problem 16.4.9

Evaluate.

$$\int_0^{2\pi} \frac{\cos\theta}{13 - 12\cos 2\theta}\, d\theta$$

Solution

The integral can be converted to a complex integral by substituting $z = e^{i\theta}$

$$\int_0^{2\pi} \frac{\cos\theta}{13 - 12\cos 2\theta}\, d\theta = \oint_C \frac{\frac{1}{2}\left(z + z^{-1}\right)}{13 - 12\left(\frac{1}{2}\left(z^2 + z^{-2}\right)\right)}\, \frac{dz}{iz} \quad = \oint_C \frac{z^2+1}{2z} \cdot \frac{z^2}{-6z^4 + 13z^2 - 6} \cdot \frac{dz}{iz} = \frac{i}{2} \oint_C \frac{z(z^2+1)}{6z^4 - 13z^2 + 6}\, dz$$

Where C is the the unit circle in the complex plane.

$f(z) = \frac{z(z^2+1)}{6z^4 - 13z^2 + 6}$ has simple poles where

$$6z^4 - 13z^2 + 6 = 0 \implies z^2 = \frac{13 \pm \sqrt{169 - 144}}{6} = \frac{13 \pm 5}{12} \implies z = \pm\sqrt{\frac{2}{3}},\ \pm\sqrt{\frac{3}{2}}$$

$$z_1 = -\sqrt{\frac{2}{3}} \quad z_2 = \sqrt{\frac{2}{3}} \quad z_3 = -\sqrt{\frac{3}{2}} \quad z_4 = \sqrt{\frac{3}{2}}$$

Only z_1 and z_2 lie within C,

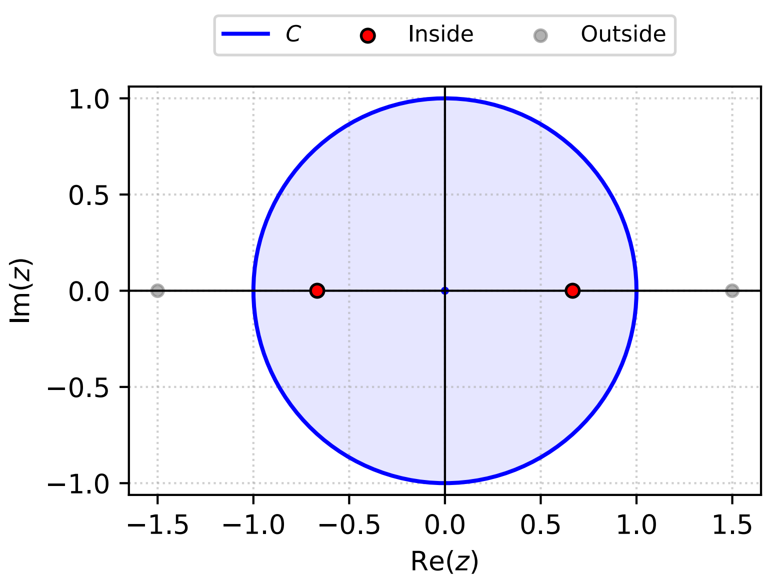

Figure 7: Poles

$$\operatorname*{Res}_{z=z_0} f(z) = \frac{z_0(z_0^2+1)}{24z_0^3 - 26z_0} = \frac{z_0^2+1}{24z_0^2 - 26}$$

$$\operatorname*{Res}_{z=z_{1,2}} f(z) = \frac{\left(\pm\sqrt{\frac{2}{3}}\right)^2 + 1}{24\left(\pm\sqrt{\frac{2}{3}}\right)^2 - 26} = -\frac{1}{6}$$

The integral is then in the form,

$$\oint_C f(z)\ dz = 2\pi i \sum_{j=1}^{k} \operatorname*{Res}_{z=z_j} f(z)$$

$$\oint_C \frac{z(z^2+1)}{6z^4 - 13z^2 + 6} = 2\pi i \left(-\frac{1}{6} - \frac{1}{6}\right) = -\frac{2\pi i}{3}$$

$$\int_0^{2\pi} \frac{\cos\theta}{13 - 12\cos 2\theta}\ d\theta = \frac{i}{2}\left(-\frac{2\pi i}{3}\right) = \boxed{\frac{\pi}{3}}$$

Problem 16.4.11

Evaluate.

$$\int_{-\infty}^{\infty} \frac{1}{(1+x^2)^2}\ dx$$

Solution

$$\int_{-\infty}^{\infty} f(x)\ dx = 2\pi i \sum \operatorname{Res} f(z)$$

$f(z) = \frac{1}{(1+z^2)^2}$ has second order poles where

$$\left(1+z^2\right)^2 = 0 \implies z = \pm i$$

Only $z = i$ lies in the upper half contour.

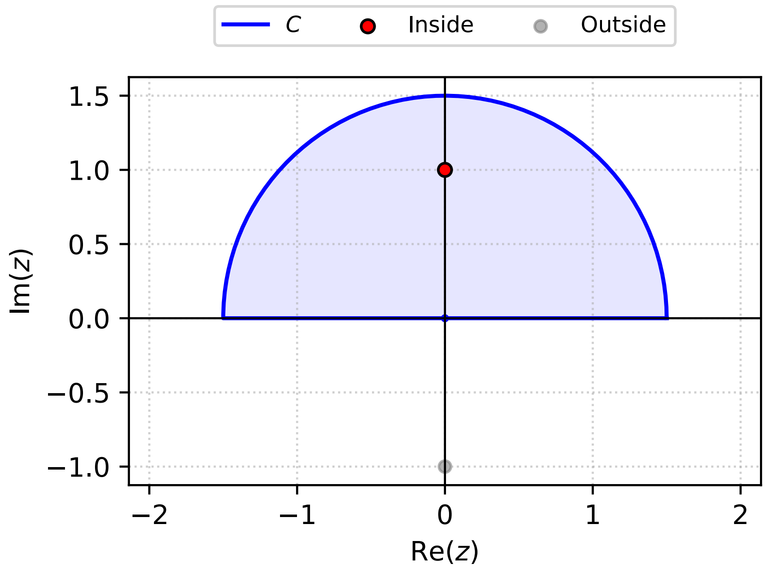

Figure 8: Poles

$$\begin{aligned}
\operatorname*{Res}_{z=i} f(z) &= \lim_{z\to i} \left\{ \frac{d}{dz} \left[\frac{(z-i)^2}{(1+z^2)^2} \right] \right\} \\
&= \lim_{z\to i} \left\{ \frac{d}{dz} \left[\frac{(z-i)^2}{(z-i)^2 (z+i)^2} \right] \right\} \\
&= \lim_{z\to i} \frac{d}{dz} \frac{1}{(z+i)^2} \\
&= \lim_{z\to i} \frac{-2}{(z+i)^3} \\
&= -\frac{i}{4}
\end{aligned}$$

$$\int_{-\infty}^{\infty} \frac{1}{\left(1+x^2\right)^2}\ dx = 2\pi i\left(-\frac{i}{4}\right) = \boxed{\frac{\pi}{2}}$$

Problem 16.4.15

Evaluate.

$$\int_{-\infty}^{\infty} \frac{x^2}{x^6+1}\, dx$$

Solution

$$\int_{-\infty}^{\infty} f(x)\, dx = 2\pi i \sum \text{Res}\, f(z)$$

$f(z) = \frac{z^2}{z^6+1}$ has simple poles where

$$z^6 + 1 = 0 \implies z_n = \exp\left\{\frac{i\pi}{6}(1+2n)\right\}, \text{ for } n = 0, 1, 2, 3, 4, 5$$

Only z_1, z_2, and z_3 lie in the upper half contour.

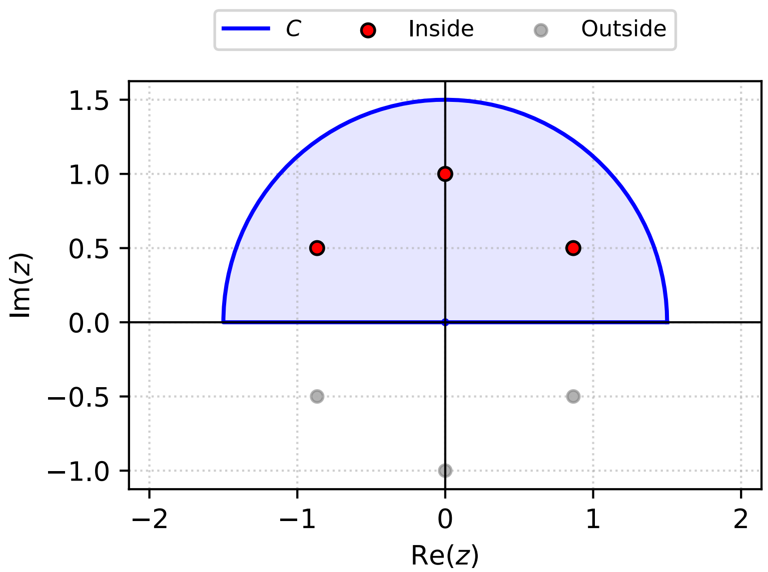

Figure 9: Poles

$$\underset{z=z_0}{\text{Res}}\, f(z) = \frac{z^2}{6z^5} = \frac{1}{6z^3}$$

$$\underset{z=z_n}{\text{Res}}\, f(z) = \frac{1}{6\exp\left\{\frac{i\pi}{6}(1+2n)\right\}^3} = \frac{1}{6}\exp\left\{-\frac{i\pi}{2}(1+2n)\right\}$$

$$\underset{z=z_0}{\text{Res}}\, f(z) = -\frac{i}{6} \quad \underset{z=z_1}{\text{Res}}\, f(z) = \frac{i}{6} \quad \underset{z=z_2}{\text{Res}}\, f(z) = -\frac{i}{6}$$

$$\int_{-\infty}^{\infty} \frac{x^2}{x^6+1}\, dx = 2\pi i\left(-\frac{i}{6} + \frac{i}{6} - \frac{i}{6}\right) = 2\pi i\left(-\frac{i}{6}\right) = \boxed{\frac{\pi}{3}}$$

Problem 16.4.17

Evaluate.

$$\int_{-\infty}^{\infty} \frac{\sin 3x}{x^4+1}\ dx$$

Solution

Problem 16.4.19

Evaluate.

$$\int_{-\infty}^{\infty} \frac{1}{x^4 - 1}\, dx$$

Solution

Problem 16.4.21

Evaluate.

$$\int_{-\infty}^{\infty} \frac{\sin x}{(x-1)\,(x^2+4)}\ dx$$

Solution

Problem 16.4.25

Find the Cauchy principal value.

$$\int_{-\infty}^{\infty} \frac{x+5}{x^3-x}\ dx$$